Gen AI

Duration: 3-6 Months | Mode: ONLINE | 100% Placement Support

Detailed Curriculum & Module Outline:

Module 1: How Language Models Work

Tokens, embeddings, context windows, temperature and why the same prompt gives different answer

Module 2: Prompt Engineering

Instructions, examples, structure and iteration - building prompts that hold up on inputs you d

Module 3: Working with the OpenAI API

Requests, streaming, structured output, function and tool calling from application code

Module 4: Open Models and Hugging Face

Finding, running and comparing open models, and the trade-offs against hosted APIs

Module 5: Embeddings and Vector Search

Chunking documents, generating embeddings, and retrieving the right context for a question

Module 6: Retrieval Augmented Generation

Building a RAG application over your own documents, with citations and honest "I do not know" b

Module 7: Fine-Tuning Basics

When fine-tuning beats prompting and retrieval, dataset preparation, and what it costs to maint

Module 8: Evaluation, Safety and Cost

Testing output quality, catching hallucinations, guarding against prompt injection and controll

Module 9: Capstone Gen AI Project

A working application built on a model, evaluated against real examples and demonstrated